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ORIGINAL ARTICLE

Neuronopathic Gaucher disease: dysregulated mRNAs and miRNAs in brain pathogenesis and effects of pharmacologic chaperone treatment in a mouse model

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Abstract

Defective lysosomal acid β-glucosidase (GCase) in Gaucher disease causes accumulation of glucosylceramide (GC) and glucosylsphingosine (GS) that distress cellular functions. To study novel pathological mechanisms in neuronopathic Gaucher disease (nGD), a mouse model (4L;C*), an analogue to subacute human nGD, was investigated for global profiles of differentially expressed brain mRNAs (DEGs) and miRNAs (DEmiRs). 4L;C* mice displayed accumulation of GC and GS, activated microglial cells, reduced number of neurons and aberrant mitochondrial function in the brain followed by deterioration in motor function. DEGs and DEmiRs were characterized from sequencing of mRNA and miRNA from cerebral cortex, brain stem, midbrain and cerebellum of 4L;C* mice. Gene ontology enrichment and pathway analysis showed preferential mitochondrial dysfunction in midbrain and uniform inflammatory response and identified novel pathways, axonal guidance signaling, synaptic transmission, eIF2 and mammalian target of rapamycin (mTOR) signaling potentially involved in nGD. Similar analyses were performed with mice treated with isofagomine (IFG), a pharmacologic chaperone for GCase. IFG treatment did not alter the GS and GC accumulation significantly but attenuated the progression of the disease and altered numerous DEmiRs and target DEGs to their respective normal levels in inflammation, mitochondrial function and axonal guidance pathways, suggesting its regulation on miRNA and the associated mRNA that underlie the neurodegeneration in nGD. These analyses demonstrate that the neurodegenerative phenotype in 4L;C* mice was associated with dysregulation of brain mRNAs and miRNAs in axonal guidance, synaptic plasticity, mitochondria function, eIF2 and mTOR signaling and inflammation and provides new insights for the nGD pathological mechanism.

Introduction

Gaucher disease is caused by mutations in *GBA1* (1,2). The resultant defects of lysosomal acid β-glucosidase (GCase) lead to accumulation of the substrates, glucosylceramide (GC) and glucosylsphingosine (GS) that affect the cellular function in visceral organs and the central nervous system (CNS) (1). Based on

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organ involvement and clinical characteristics, Gaucher disease variants are classified into three types. Type 1 presents visceral manifestations including hepatosplenomegaly and anemia (1,2). Type 2 is an acute progressive neuropathic variant (2–4), and type 3 is a progressive subacute neuropathic variant with visceral involvement (1,5). Types 2 and 3 will be abbreviated as neuronopathic Gaucher disease (nGD) as they represent a continuum of CNS disease. Gaucher disease pathogenesis in the CNS is associated with the toxic effects of GC and GS and their impact on neuronal degeneration (5,6). The molecular pathways underlie neurodegeneration in nGD has been elusive.

To understand the pathophysiology and to facilitate the development of therapeutic approaches for the nGD, several mouse models were generated including the genetic knock out of *Gba1*, conditional knock out of *Gba1* and a viable neuronopathic mouse model (4L;C*) (7–10). The 4L;C* strain was developed by cross-breeding of V394L GCase, a known human nGD allele homozygote (11) into the isolated saposin C-deficient mouse (12). The resultant 4L;C* mice showed GC and GS accumulation and neurologic phenotype specifically in the CNS and faithfully mimics nGD phenotype in human (10,13). The 4L;C* mice survive for ~48 days and die from progressive CNS disease and substrate accumulation. The V394L homozygote mice themselves do not develop substrate accumulation and progressive CNS disease, and saposin C-deficient mice do not manifest a CNS phenotype and excess substrates before 12 months (11,12,14). Therefore, the saposin C-deficient impairments will not interfere with the phenotypic or biochemical studies of nGD 4L;C* model. The 4L;C* mice were treated with isofagomine (IFG), a potent GCase reversible competitive inhibitor and an effective *in vitro* chaperone, that enhanced V394L GCase activity by stabilizing selected GCase mutant proteins and facilitating their trafficking to lysosome (15–17). IFG treatment did not alter the GS and GC accumulation significantly but slowed CNS disease progression and suppressed CNS inflammation in 4L;C* mice (16). Despite these *in vivo* effects of IFG, the mechanism for its *in vivo* effects on the nGD neurodegeneration remains poorly understood.

Profound systematic and CNS pathophysiological changes in Gaucher disease implicate very complex interactions at molecular, cellular, histological and organismal levels during disease course (1). Microarray and next generation sequencing technologies were used to explore the transcriptomes in mice with *Gba1* mutations and in prosaposin-deficient mice and provided insights into the molecular events underlying glycosphingolipid storage diseases (18–21). These studies have shown correlations between neuropathic involvement and gene expression in brains from nGD patients or *Gba1* variant mice (20,21). However, there has been no study thus far to identify the region-specific involvement of the miRNA and mRNA and the associated biological pathways in nGD brain. Understanding the fundamental features of miRNA and their correlation to mRNA profile in the brain of Gaucher disease would further delineate new aspects of the disease pathogenesis and potentially have the clinical implications in nGD.

In this study, the substrate accumulation, sensorimotor function, neurons and microglial cells status and mitochondrial function were characterized in 4L;C* mice and region-specific transcriptomic changes of brain mRNAs and miRNAs were analyzed in 4L;C* brain regions. IFG-treated 4L;C* mice were analyzed to identify the miRNAs, mRNAs and pathways normalized in the 4L;C* mice post-treatment. These analyses demonstrated that 4L;C* mice displayed deteriorated CNS function with increased occurrence of inflammatory subsets of microglial cells and loss of neurons. The expression changes in mRNAs and miRNAs were shown to relate to synaptic plasticity, axonal guidance, mitochondrial function, eIF2 and mTOR signaling and CNS inflammation that underlie the neurodegenerative phenotypes.

Results

Neuronopathic Gaucher disease mouse model, 4L;C* mice

Phenotype

4L;C* mice were born without phenotypic abnormalities. Neuronal abnormalities begin at the age of ~30 days and death occurs by ~48 days (10). The progression of neurological deficits was quantitated by gait analyses of sensorimotor function. The right and left strides and the base width were significantly abnormal and progressive after 35 days of age (Fig. 1A). Suppression of long-term potentiation (LTP) in hippocampal slices from 35-day 4L;C* brains was evident as determined in previous study indicating defects of synaptic plasticity (10).

Brain substrate levels and mitochondrial function

The 4L;C* mice had excess GC and GS levels in the whole brain extracts (10). Here, the respective GC concentrations in cortex (CO), brain stem (BS), midbrain (MID) and cerebellum (CB) regions were 4.4-, 9.0-, 6.5- and 8.1-fold greater than those in the corresponding wild-type (WT) control regions (Fig. 1B). Similarly, relative to WT regions, the GS concentrations were greatly increased in CO, BS, MID and CB from 4L;C* mice; GS was barely detectable in WT controls (Fig. 1B). The GC and GS levels in the brains of age-matched V394L homozygote and saposin C-deficient mice are at the normal ranges (11,12,14). Isolated mitochondria from 4L;C* brains showed reduced oxygen consumption and ATP production compared with WT levels, indicating defective mitochondrial function (Fig. 1C).

Characterization of microglial cells and neurons in 4L;C mouse brains*

The inflammatory subsets of microglial cells and the degree of neuronal loss in the 4L;C* brains were determined in single cell suspensions prepared from strain-matched 4L;C* and WT brains. Map2 is a neuron marker. As compared with WT, the 4L;C* brains showed significantly reduced $CD11b^{-}$ $CD45^{-}$ and MAP2+ neuronal cells as a percentage of total cells (~75%) and on a peak shift of the flow cytometry histogram (Fig. 1D). The reduced number of Map2-positive cells indicates the neuron loss, a neurodegenerative feature in 4L;C* mice. The regional neuron loss was determined by immunohistochemistry on WT and 4L;C* brain sections with antibody to Map2. These data showed the decreases of Map2 signal intensity in the brain with pronounced effect on MID and CB of 4L;C* brain when compared with WT mice (Fig. 1E).

In comparison, the percentages of $CD45^{int}CD11b^{+}CX3CR1^{+}$ microglial cells were significantly higher (~2.5-fold) in 4L;C* mice when compared with WT (Fig. 1F and G). These microglial cells showed increased protein expression of CCR2, Ly6C and Ly6G, which are inflammatory markers expressed by systemic macrophages (Fig. 1H). The strong positive signal of these microglial cells for CCR2, Ly6G and Ly6C suggested that monocytes were preferentially recruited to the lesions in the brain and differentiated into macrophages-like cells in 4L;C* brain. Activated microglial/macrophage in 4L;C* brain stained positive for anti-CD68 antibody and showed uniform inflammatory response across brain regions (Fig. 1I).

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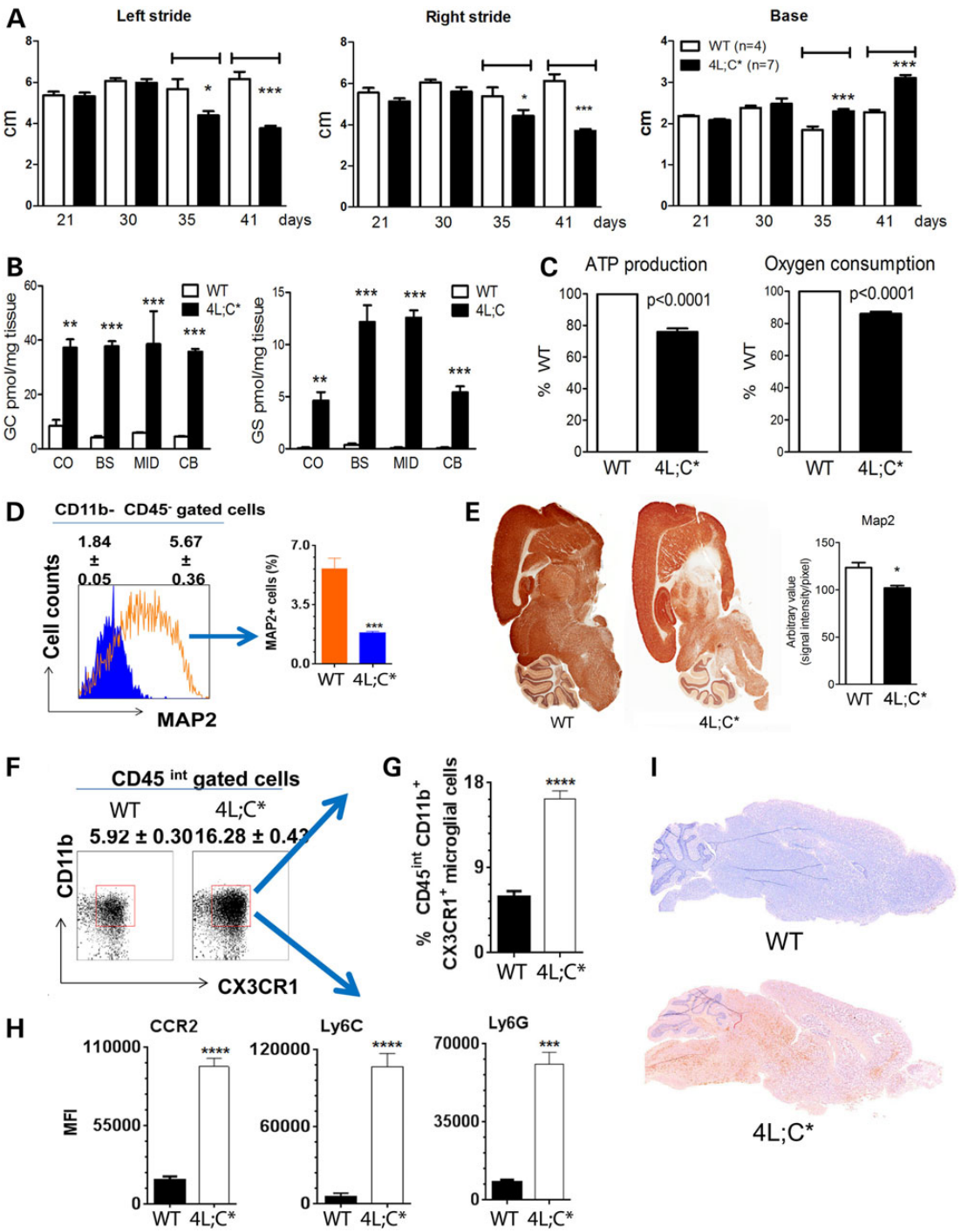


Figure 1. Phenotypes of 4L;C* mice. (**A**) Gait analysis. 4L;C* mice showed significant decrease on left and right strides and increase on base width at 35 days of age and older. The tests repeat two to three times ($n = 4$–7 mice). (**B**) Glucosylceramides (GC) and glucosylsphingosine (GS) quantitation by LC/MS. Both GC (left panel) and GS (right panel) levels were significantly increased in 4L;C* brain regions at 45 days of age relative to WT levels ($n = 3$ mice). (**C**) Mitochondrial function. Percentage of ATP production and oxygen consumption rate were decreased in 4L;C* brain compared with WT mice. The data are from duplicate experiments ($n = 2$ mice). (**D**) Augmented loss of neurons in 4L;C* mice. Single-cell suspensions prepared from whole brain of 4L;C* and WT mice were analyzed by FACS. These cells were first gated for negativity of antibodies to CD11b and CD45 and then stained for positivity of antibodies to Map2, a neuron marker. The histogram (left) and their corresponding columns (right) showed decreased percentages of $MAP2^+$ cells in 4L;C* mice brain (blue) compared with WT mice (orange) ($n = 10$ mice). (**E**) Immunohistochemistry of Map2 antibody staining on 4L;C* and WT brain. Map2 signals (brown) in 4L;C* brain were significantly decreased compared with WT brain at age of 45 days. Signal intensity was quantitated by NIH image J software ($n = 2$ mice, 2 sections/mouse). (**F–H**) Increased incidence of inflammatory subsets of microglial cells in 4L;C* mice brain. Single cell suspensions from the whole brain of 4L;C* and WT mice were analyzed by FACS. FACS-sorted CD45int CD11b+ and CX3CR1+ microglial cells (F) shown as percentage were significantly increased in 4L;C* compared with WT brain (G). Microglial cells have CD45int CD11b+ and CX3CR1+ positivity stained for antibodies to CCR2, Ly6C and Ly6G to evaluate their mean fluorescence intensity (MFI). 4L;C* brain had increased signals for CCR2, Ly6C and Ly6G than WT brains (H). Two independent experiments were conducted ($n = 10$ mice). (**I**) Inflammation in 4L;C* brain was determined by anti-CD68 (a marker for activated microglia/macrophage) antibody staining (brown). WT mice brain showed background level. Hematoxylin stains cell nuclei (blue). The values are represented as mean ± S.E. Group comparison was done by Student's *t*-test (*$P < 0.05$; ***$P < 0.001$; ****$P < 0.0001$).

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IFG treatment on 4L;C* mice

The effects of IFG treatment on the 4L;C* phenotype were reported previously (16). IFG-treated 4L;C* mice had reduced inflammatory signals in the brain and significant extension of life span, up to 67 days with median survival of 63 days, relative to untreated mice that died at about 48 days (Table 1). The reduced inflammatory responses in IFG-treated 4L;C* brain were demonstrated by significant reduction of the macrophage marker, CD68, astrocyte marker, GFAP and the inflammation indicator, TNFα (16). Although GCase activity and protein levels were increased in the treated mice, IFG treatment did not significantly reduce brain GC or GS levels in 4L;C* mice owing to IFG as a competitive inhibitors of GCase (16,17) but still attenuated the neuronal phenotype including inflammation and other potential pathological targets. These findings suggest that the therapeutic effects of IFG were mediated by anti-inflammatory and other biological effects in CNS. The transcriptome analyses of IFG-treated 4L;C* brain in correlation with untreated brains will identify the biological pathways normalized by IFG. Important to these studies, improved molecular changes in IFG samples will serve as significant reference to validate the molecular findings in 4L;C* mice.

Biological function analyses of differentially expressed mRNAs

To investigate the disease-related molecular processes in the brain of Gaucher disease, transcriptome analyses of mRNAs were performed on extracts from the different brain regions of the 4L;C* mice. The effect of IFG treatment on the regional expression of mRNAs in the brain was also evaluated to identify the genes or pathways involved in the improvement of the disease processes.

Profiling differentially expressed mRNAs in 4L;C brain regions*

Differentially expressed mRNAs (DEGs) were identified in all four brain regions in the 4L;C* mice. In comparison with the WT brain, 4L;C* mice BS had the highest number of DEGs followed by the CO and MID. CB had the least number of DEGs with ~50% fewer than the other regions. The DEGs were stratified into up and down regulated groups (Fig. 2A). Venn diagrams showed 647 DEGs that overlapped (common) between the four brain regions as well as region-specific DEGs in CO (939), BS (1230), MID (1044) and CB (276), respectively (Fig. 2B). The gene list is shown in Supplementary Material, Table S1.

Table 1. IFG treatment on 4L;C* mice

		4L;C*	4L;C*+IFG[a]	P[b]
GCase	Activity in midbrain (fold change to 4L;C*)	1	1.4	<0.05
Substrates (pmol./mg protein)	GC in cortex	420	400	ns
	GS in cortex	20	20	ns
Proinflammation in brain	CD68[c]	High	Low	<0.02
	GFAP[c,d]	High	Low	<0.01
	TNFα[e] (fold change to WT)	35	17	<0.05
	IL6[e] (fold change to WT)	13	9	ns
	Phospho-p38[d]	High	Low	<0.0001
Life span	Median survival days	48	63	<0.0001[f]

[a]The data of the IFG treatment on 4L;C* mice shown in this table were described previously (16).
[b]Student's *t*-test; ns, not significant.
[c]CD68 and GFAP signal level was determined by immunohistochemistry.
[d]GFAP and phospho-p38 levels were determined by immunoblot.
[e]TNFα and IL6 levels were determined by qRT-PCR.
[f]Log-rank (Mantel-Cox) test.

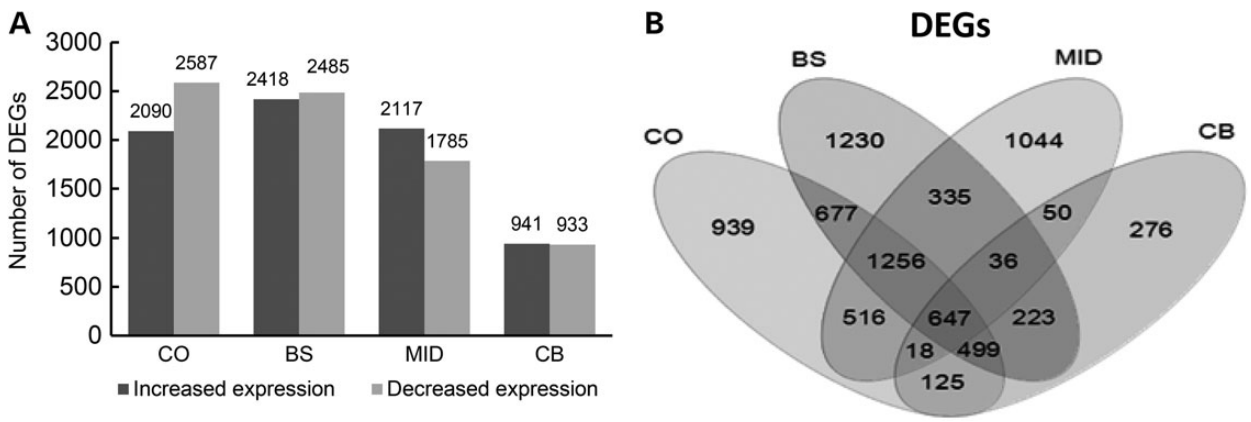


Figure 2. Differentially expressed mRNA in 4L;C* brain. DEGs in 4L;C* brain regions, CO, BS, MID and CB, were determined by RNASeq analyses. (**A**) The number of DEGs with increased expression (dark gray bars) or decreased expression (light gray bars) is labeled on the top of bars, respectively. (**B**) Venn diagram analyses show the commonality of DEGs (in A) derived from four different brain regions as indicated, respectively.

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Inflammatory versus non-inflammatory DEGs

Highly expressed CNS inflammatory response genes in 4L;C* brain could mask the genes involved in other neurological function. The DEGs were first segregated into inflammatory and non-inflammatory groups (others) (Fig. 3), and then the canonical pathways and functions of the DEGs were analyzed by ingenuity pathway analyses (IPA) (Tables 2 and 3). The distribution of inflammatory versus non-inflammatory DEGs was almost similar in each 4L;C* brain region: inflammatory DEGs being ~17–26% of the total DEGs (Fig. 3A), suggesting a generalized inflammatory responses across 4L;C* brain regions (Supplementary Material, Table S2). Approximately 80% of the remaining DEGs in the different brain regions were non-inflammatory, indicating large non-inflammatory components involved in the pathogenic mechanisms. IFG treatment reduced the number of DEGs in inflammatory and non-inflammatory categories across 4L;C* brain regions (Fig. 3B and C), indicating non-inflammatory gene expression was also affected by IFG. The DEGs in non-inflammatory category were used for eIF2, mTOR, mitochondrial function, axonal guidance and synaptic transmission pathway analyses.

Inflammatory DEGs

Inflammatory response is a neurodegenerative feature in 4L;C* brains (Fig. 1) (10). Significantly enriched canonical pathways were identified from the inflammatory DEGs in both untreated and IFG-treated 4L;C* mice. The pathways include those with

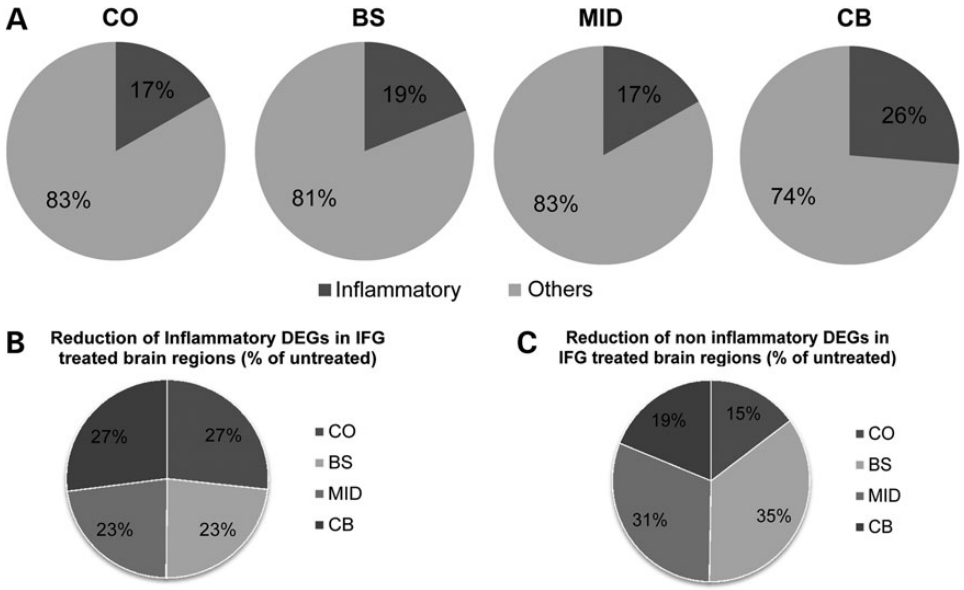


Figure 3. Category of inflammatory and non-inflammatory DEGs. (**A**) Proportion of inflammatory DEGs in each brain region. The number of inflammatory DEGs distributes uniformly across four brain regions in 4L;C* mice. (**B**) Reduction of inflammatory DEG numbers (%) in IFG-treated 4L;C* brain regions. (**C**) Reduction of non-inflammatory (other) DEG numbers (%) in IFG-treated 4L;C* brain regions.

Table 2. Number of DEGs in canonical pathways of inflammatory DEGs in untreated (UT) and IFG-treated (T) 4L;C* brain regions

Ingenuity canonical pathways	CO		BS		MID		CB	
	UT	T	UT	T	UT	T	UT	T
Role of macrophages, fibroblasts and endothelial cells in rheumatoid arthritis	51	32	81	47	43	26	43	28
Dendritic cell maturation	48	31	55	40	38	29	32	22
Acute phase response signaling	43	28	48	36	34	35	31	14
Granulocyte adhesion and diapedesis	43	23	49	43	35	25	37	21
NF-κB signaling	41	28	55	27	29	18	26	18
Agranulocyte adhesion and diapedesis	39	24	50	42	40	27	33	19
Production of nitric oxide and reactive oxygen species in macrophages	38	30	41	27	31	21	24	15
Role of NFAT in regulation of the immune response	36	25	49	32	31	17	25	15
Altered T-cell and B-cell signaling in rheumatoid arthritis	33	14	36	26	25	13	24	12
LXR/RXR activation	33	19	33	24	26	24	21	11
Role of pattern recognition receptors in recognition of bacteria and viruses	32	17	39	30	23	20	23	18
IL-12 signaling and production in macrophages	32	24	30	19	23	16	14	10
Communication between innate and adaptive immune cells	31	11	32	20	25	15	20	12
PKCθ signaling in T lymphocytes	31	24	39	22	20	11	20	10
Crosstalk between dendritic cells and natural killer cells	30	10	35	20	18	13	20	9
TNFR1 and TNF2 signaling	11	8	18	8	6	3	6	4
JNK signaling	8	7	10	5	7	3	2	3
IL4 signaling	7	4	8	5	6	4	5	2
IFγ signaling	15	11	18	12	13	9	8	6

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Table 3. Number of non-inflammatory DEGs in top canonical pathways and functions and disease groups in untreated (UT) and IFG-treated (T) 4L;C* brain regions

Category	CO		BS		MID		CB	
	UT	T	UT	T	UT	T	UT	T
Neurological disease	64	42	174		254		29	
Molecular transport	16	13	169		52		32	
eIF2 signaling	75	60	84	6	91	16	0	1
Regulation of eIF4 and p70S6K signaling	46	32	46		48		46	
Axonal guidance signaling	30	20	34	12	26	13	11	14
mTOR signaling	45	31	56	10	52	11	6	5
CREB signaling in neurons	24	17	30	12	28	14	7	
Mitochondrial dysfunction	70	31	29	4	76	19	9	7
RNA post-transcriptional modification	34	40	68		78		27	
Metabolic disease	34	23	57		40		25	
Dopamine receptor signaling	7	6	18	2	14	3	3	2
Glutamate receptor signaling	7	3	17	9	10	5	3	7
GABA receptor signaling	5	6	12	7	7	3	2	1
Lipid metabolism	14	16	28		20		31	
Cell death and survival	4		10		21		37	

roles in macrophage and dendritic cell maturation, NF-κB signaling and TNF signaling (Table 2). There was no region-specific enrichment of these functional pathways, suggesting similar inflammatory reaction across the 4L;C* brain. IFG treatment reduced the number of inflammatory DEGs by 73–77% in the four brain regions (Fig. 3B), affecting pro-inflammatory (Ifnγ and Tnf) and anti-inflammatory (Il4) signaling pathways (Table 2 and Supplementary Material, Table S2). Decreased *Tnfα* expression found previously in IFG-treated 4L;C* mouse brains (Table 1) (16) was evident even in transcriptomic analysis (Supplementary Material, Table S2). The results of inflammatory DEG analyses indicate a relatively uniform inflammatory response across 4L;C* brain regions with both pro-inflammatory and anti-inflammatory genes being altered or suppressed by IFG in 4L;C* brain.

DEGs in eIF2 signaling pathway

eIF2 signaling is fundamental for translation initiation and protein synthesis. A total of 86 DEGs related to eIF2 pathway were found in the 4L;C* brain regions, and all showed increased expression (Fig. 4A). DEGs in this pathway consist of ribosome proteins (79%) and various eukaryotic initiation factors (15%) (Fig. 4B and Supplementary Material, Table S3). This suggests a major involvement of the translational machinery in the pathogenesis in 4L;C* brain. In IFG-treated 4L;C* mice, a number of eIF2 pathway DEGs were reduced (Fig. 4B and Supplementary Material, Fig. S1 and Table S3). Based on the number of DEGs, the genes in eIF2 signaling pathway in BS appeared more sensitive than other brain regions in response to IFG. These analyses demonstrated that genes in eIF2 signaling pathways were upregulated in 4L;C* brains and markedly affected by IFG.

DEGs in mammalian target of rapamycin (mTOR) signaling pathway

mTOR signaling pathway serves as a central regulator of the cell metabolism, growth, proliferation and survival in response to intracellular or extracellular signals (growth factors, hormones, mitogens and cytokines). A total of 57 DEGs in this pathway were identified, all showed an increased expression (Fig. 4C and Supplementary Material, Table S3). Most of the DEGs were ribosomal proteins in this pathway (Fig. 4D), which indicates a tightly functional interaction of mTOR signaling pathway with eIF2 pathway. IFG treatment reduced the number of mTOR pathway DEGs (Fig. 4C and Supplementary Material, Fig. S2 and Table S3). Altered gene expression in mTOR signaling pathway indicates the involvement of this pathway in nGD pathogenesis. IFG treatment had effect on preventing the alteration of mTOR genes expression.

DEGs in mitochondrial function

Nuclear-encoded DEGs related to mitochondrial functions were identified in 4L;C* brain regions, and these showed regional expression patterns (Fig. 5 and Supplementary Material, Table S3). The expression profile of mitochondrial genes is shown in hierarchical cluster analyses (Fig. 5A). Interestingly, MID had a distinct pattern of increased DEGs compared with other regions (Fig. 5A and B). Among the 115 DEGs associated with mitochondrial function, 34 belonged to NADH dehydrogenase (ubiquinone) subcomplex, 16 of them were components of cytochrome c oxidase subunit, 11 were part of ATP synthase and H+ transporting mitochondrial complex subunit and 5 ubiquinol-cytochrome c reductase were found (Supplementary Material, Table S4). These nuclear mitochondrial genes encode proteins for components of complexes I–V (22). Most of those genes had enhanced expression in 4L;C* brain, with greater increases in MID and CO, followed by BS and CB. The increased expression of those nuclear mitochondrial genes suggests a compensatory mechanism to restore defective mitochondrial function in the neurodegeneration of the 4L;C* brain.

Additional DEGs in mitochondria function group include *Lrrk2* and *Park2*. *Lrrk2* is the gene encoding Leucine-Rich Repeat Kinase 2. *Park2* encodes the E3 ubiquitin ligase parkin. The mutations on *Lrrk2* and *Park2* have been associated with Parkinson's disease (PD) (23,24). The expression of *Lrrk2* in CO, BS and MID and *Park2* in CO, BS and CB were decreased in 4L;C* brain (Supplementary Material, Table S3). The altered expression of these PD-related genes could be associated with the abnormal α-synuclein pathology in nGD brains (25,26).

IFG treatment considerably reduced the mitochondrial DEGs in BS and MID regions, but barely on CO and CB regions of 4L;C* mice (Fig. 5C). IFG did not regulate the expression of all the genes, e.g. altered expression of *Park2* was not normalized by IFG treatment (Supplementary Material, Table S3). The analysis suggests that MID and BS regions are sensitive to mitochondrial

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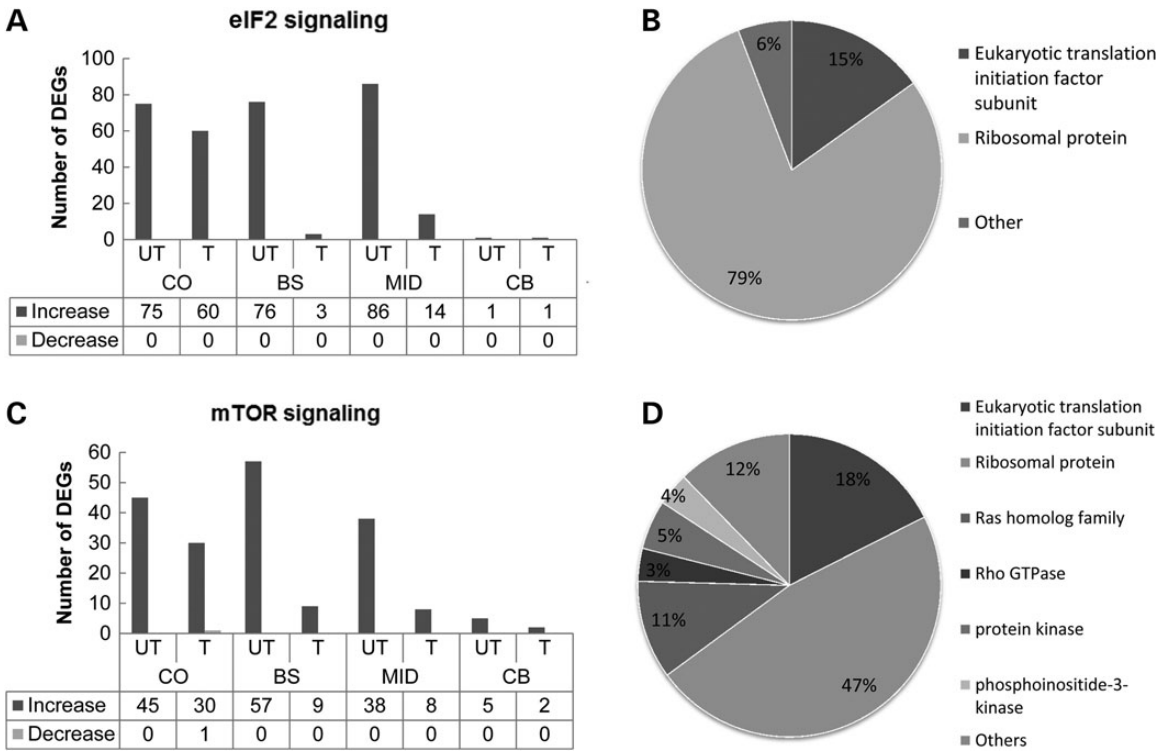


Figure 4. eIF2 and mTOR signaling pathways. (**A**) Number of eIF2 DEGs. DEGs were enriched in CO, BS and MID. Most of the DEGs had increased expression. (**B**) Functional category and their fraction of eIF2 DEGs. (**C**) Number of mTOR DEGs. DEGs were enriched in CO, BS and MID. Most of the DEGs had increased expression. (**D**) Functional category and their fraction of mTOR DEGs. The number of DEGs in eIF2 and mTOR pathways was reduced in IFG-treated 4L;C* brain regions. UT, untreated; T, IFG treated.

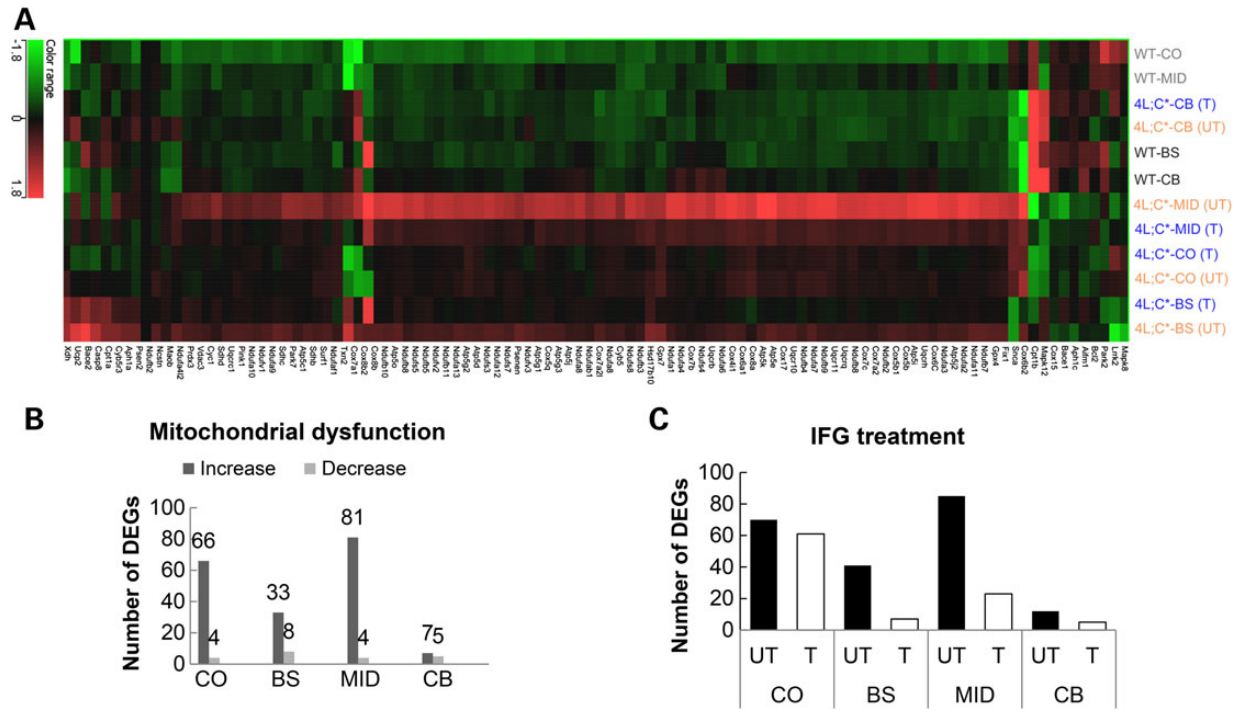


Figure 5. Mitochondrial function. (**A**) Hierarchical cluster analyses of 106 mitochondrial DEGs in WT, treated (T) 4L;C* and untreated (UT) 4L;C brain in CO, BS, MID and CB region. Each row represents a single DEG across four brain regions. Hierarchical charts are clustered by normalized intensity values (green-to-red scale bar) representing the fold changes (−1.8 to +1.8) of DEG. (**B**) Number of mitochondrial DEGs. MID had greater number of DEG than that in CO, BS and CB. Most of the DEG had increased expression. (**C**) Reduction of DEG numbers in IFG-treated 4L;C* brain regions. Untreated (UT, black bars) and treated (T, open bars).

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stress in Gaucher disease. IFG treatment had normalization effects on some of the mitochondrial genes involved in respiration chain function.

DEGs in axonal guidance signaling

The conserved families of guidance cues and their neuronal receptors generate axonal guidance signals to allow precise path finding of neuronal connectivity for nervous system development and repair. Most of the DEGs in axonal guidance signaling pathway had decreased expression in 4L;C* brain regions (Fig. 6A and Supplementary Material, Table S4). The affected genes (circled) include the receptors in signaling pathways for axonal repulsion, attraction and growth (Fig. 6B). Additional DEGs included: *Arhgef11* that is involved in G proteins and Rho-dependent signals (27); FZDs, the receptors for Wnt signaling and axon growth (28); and ADAM metalloproteases that have roles in the regulation of axonal guidance (29) (Supplementary Material, Table S4). The altered gene expression in this pathway suggests the impaired axonal growth and regeneration involved in the pathogenesis in 4L;C* brain. In the IFG-treated 4L:C* brains, the numbers of axonal guidance DEGs were reduced in CO, BS and MID, but increased in CB (Fig. 6C and Supplementary Material, Table S4). About 50% of DEGs were normalized or reduced to their expected normal levels in the IFG-treated 4L;C* mice, indicating partial correction of abnormal gene expression in this pathway by IFG.

DEGs in synaptic transmission/LTP

Decreased LTP in 4L;C* brain indicates defective synaptic transmission. In 4L;C* brain, the expression profile of the DEGs involved in the synaptic transmission was not uniform with both increased and decreased expression patterns (Fig. 7A and Supplementary Material, Table S4). They include the N-methyl-D-aspartate (NMDA) and α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid glutamate receptors, ryanodine receptors (Ryr) and ion channels. Most of the DEGs involved in LTP as grouped by IPA had decreased expression (Fig. 7B and Supplementary Material, Table S4). These included the Ryrs (*Ryr1*, *Ryr2* and *Ryr3*), which were decreased in all four regions (Fig. 7C and Supplementary Material, Table S4). The Ryr family modulates cellular calcium concentration, and its deficiency could affect synaptic plasticity and many cellular functions including calcium homeostasis, mitochondria and proteases. In addition to *Ryrs*, other DEGs included the calcium channels and calcium binding proteins that may play the role in calcium homeostasis (Fig. 7D and Supplementary Material, Tables S1 and S4). As calcium homeostasis is essential for maintaining LTP, decreased expression of these LTP related genes could promote the suppression of LTP in 4L;C* mice (10). In IFG-treated 4L;C* brain, the number of DEGs in the synaptic transmission/LTP functional groups was reduced by ~20–50% in MID, BS and CO regions (Fig. 7A and B). However, the DEGs involved in LTP, e.g. *Ryrs* (Fig. 7C) and *Cacna1d* and *Cacna1g* (Fig. 7D), were not corrected by IFG treatment.

Analyses of DEmiRs

To explore the disease-related miRNAs in nGD, miRNAs in the 4L;C* mice brain and effect of IFG on DEmiRs was evaluated to identify the target DEGs and the pathways involved in the disease processes.

DEmiRs in 4L;C brain regions*

The miRNA were isolated from the identical brain samples as used for DEGs identification. Unlike the DEGs, the DEmiRs were highest in number in MID, whereas CO, BS and CB had similar numbers of DEmiRs (Fig. 8A and Supplementary Material,

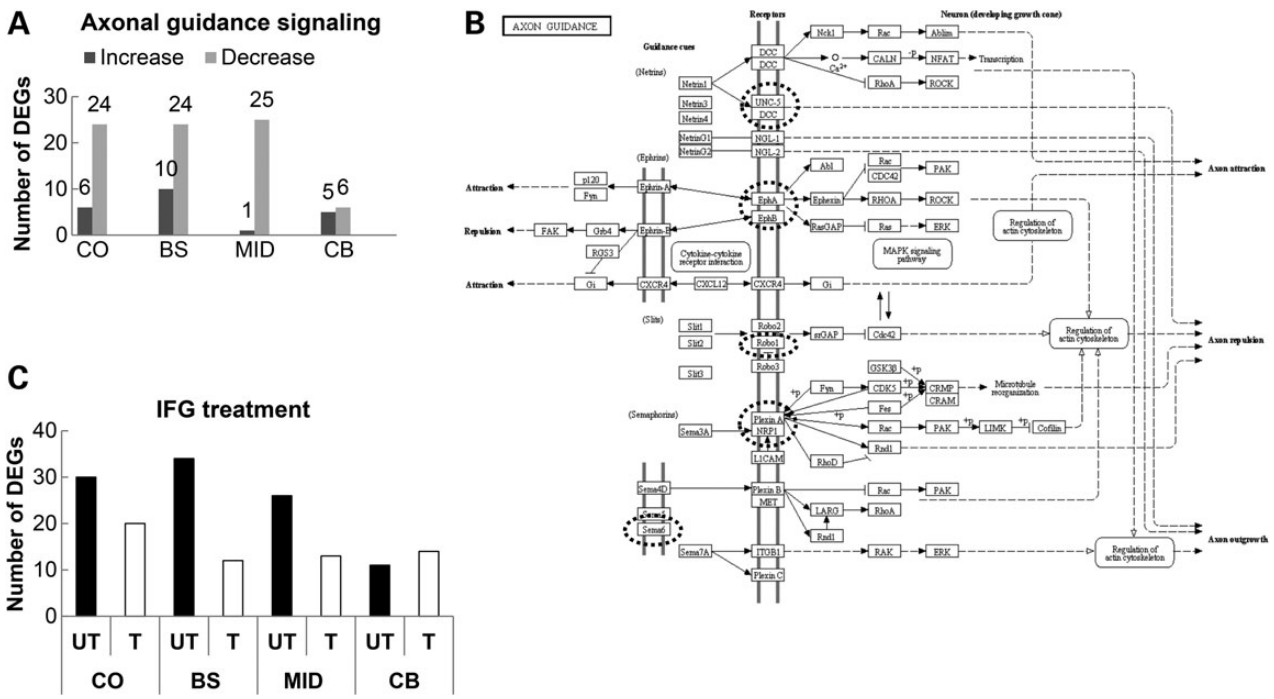
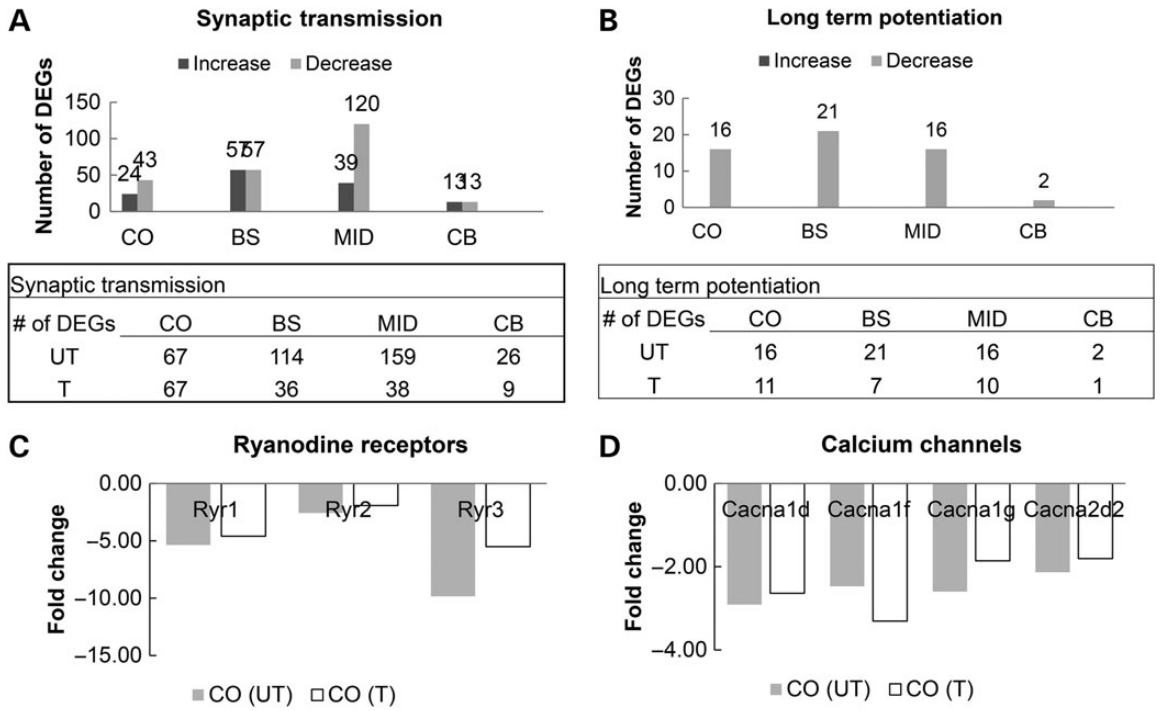


Figure 6. Axonal guidance signaling. (**A**) Number of axonal guidance DEGs. Most of the DEGs in CO, BS and MID showed decreased expression. (**B**) Function of DEGs in axonal guidance. The DEGs involved were highlighted with dotted circles in the pathway. The axonal guidance graph adapted from KEGG (download of dated 9 June 2014). (**C**) IFG treatment on axonal guidance DEGs. The number of DEGs in 4L;C* brain regions was reduced in IFG-treated CO, BS and MID. Untreated (UT, black bars) and treated (T, open bars).

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Synaptic transmission				
# of DEGs	CO	BS	MID	CB
UT	67	114	159	26
T	67	36	38	9

Long term potentiation				
# of DEGs	CO	BS	MID	CB
UT	16	21	16	2
T	11	7	10	1

Figure 7. DEGs for synaptic transmission and LTP. (**A**) Number of DEGs for synaptic transmission. (**B**) Number of DEGs for LTP. Number of DEGs in IFG treated (T) compared with untreated (UT) brain regions for synaptic transmission and LTP are shown in the table below (A or B), respectively. (**C**) Decreased expression of *Ryrs* in both untreated (UT) and IFG-treated (T) CO. (**D**) Decreased expression of calcium channels in both untreated (UT) and IFG-treated (T) CO. The expression levels are plotted as fold change relative to WT level in (C) and (D).

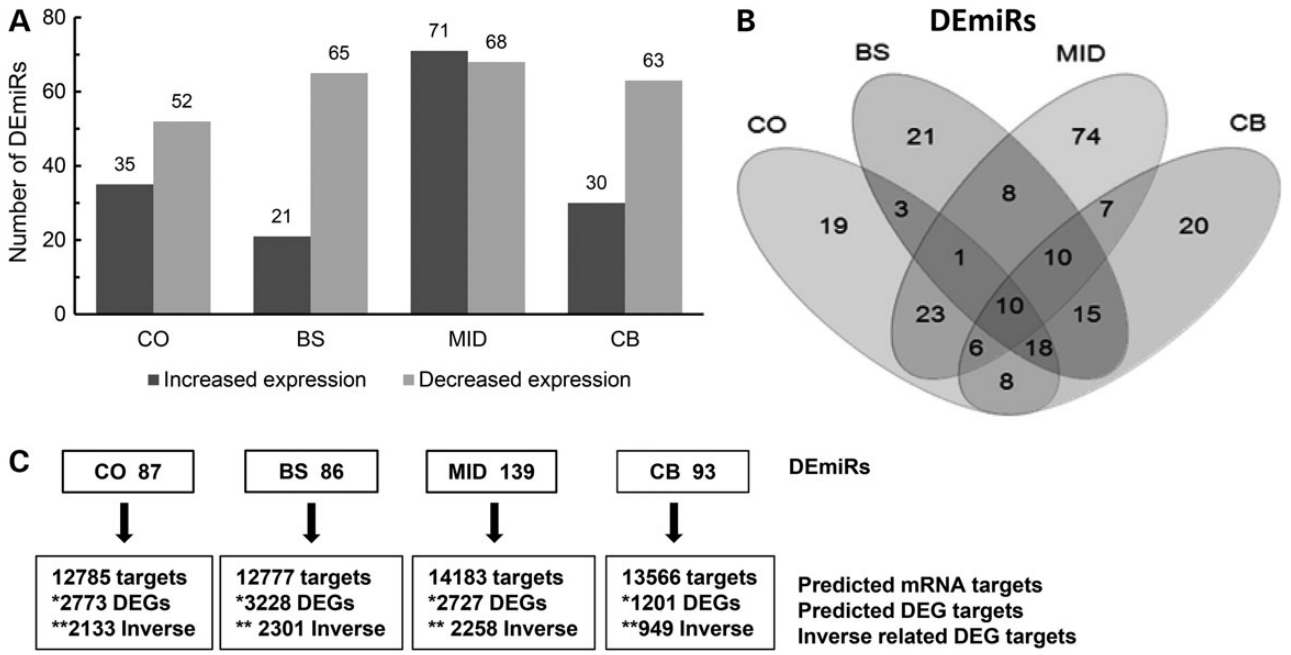


Figure 8. Differentially expressed miRNA in 4L;C* brain. DEmiRs in 4LC brain regions (CO, BS, MID and CB) were determined by RNASeq analyses. (**A**) The number of DEmiRs with increased expression (dark gray bars) or decreased expression (light gray bars) is labeled on the top of bars, respectively. (**B**) Venn diagram analyses show the commonality DEmiRs (in A) derived from four different brain regions as indicated, respectively. (**C**) DEmiRs and predicted DEG targets in 4L;C* brain regions.

Table S5). Venn diagrams showed 19, 21, 74 and 20 region-specific DEmiRs in CO, BS, MID and CB, respectively, and 10 were common to all 4 regions (Fig. 8B). IFG treatment reversed the expression of multiple DEmiRs and the associated mRNAs to normal levels (Fig. 9). Compared with untreated 4L;C* mice, DEmiR numbers were reduced by 40–60% in the IFG-treated 4L;C* brain (Fig. 9A

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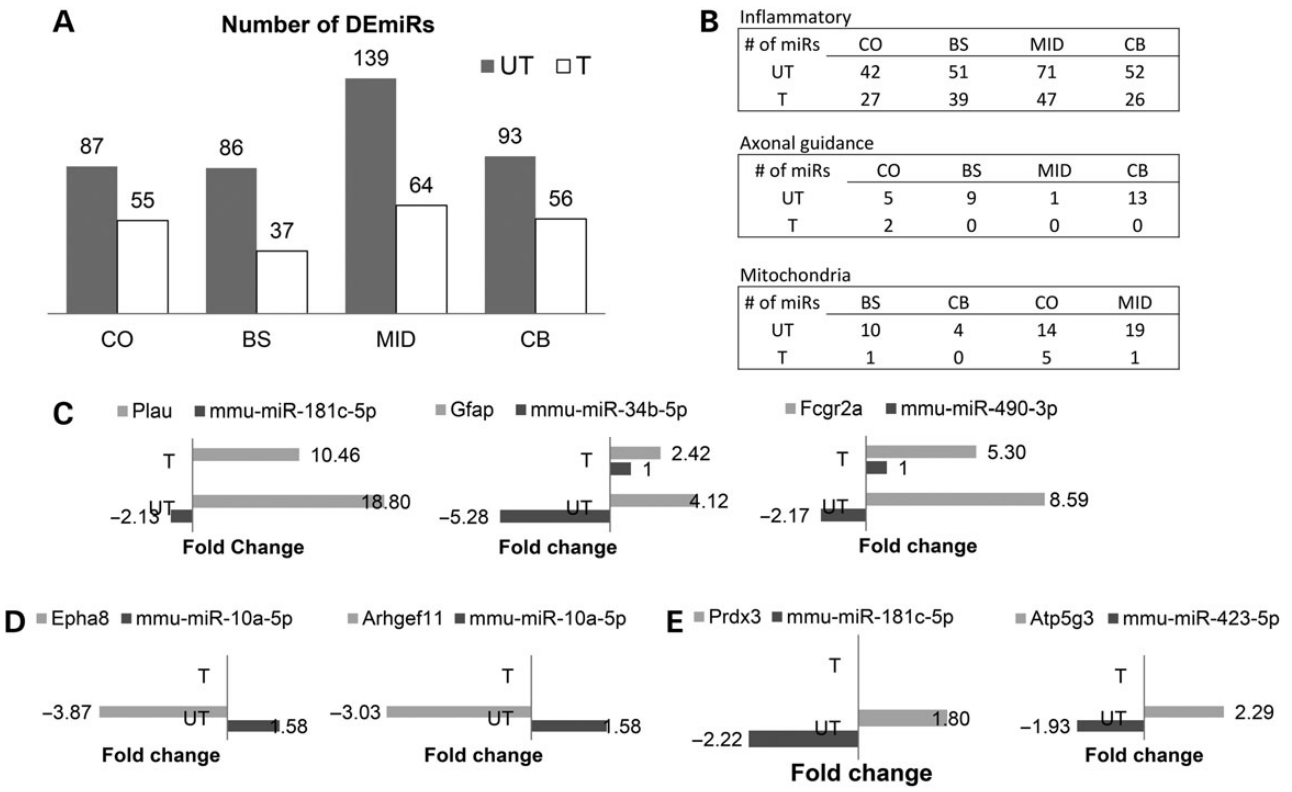


Figure 9. Dysregulated miRNAs in 4L;C* brain. (**A**) Number of DEmiRs in IFG-treated (open bar, T) and untreated (gray bar, UT) 4L;C* brain regions. (**B**) Number of DEmiRs in functional pathways of IFG-treated (T) and untreated (UT) 4L;C* brain regions. (**C**) DEmiRs in inflammatory group and IFG effect. Decreased miR-181c-5p, miR-34b-5p and miR-490-3p correlated with increased expression of inflammatory DEG targets, *Plau*, *Gfap* or *Fcgr2a*, respectively, in untreated (UT) 4L;C* BS. IFG treatment (T) partially normalized their expression. (**D**) DEmiRs in axonal guidance signaling pathway and IFG effect. In IFG-treated (T) 4L;C* MID, both miR-10a-5p and its target DEG, *Epha8* and *Arhgef11*, were changed to normal level compared with untreated (UT). (**E**) DEmiRs in mitochondrial group and IFG effect. Compared with untreated (UT), both miR-423-5p/*Atp5g3* and miR-181c-5p/*Prdx3* were changed to normal level in treated MID (T).

and Supplementary Material, Table S5), suggesting potential effect of IFG on miRNAs expression.

Prediction of miRNA-targeted mRNA

Target scan software was used to identify the miRNAs that regulate gene/target mRNA expression (30). Target mRNAs of each DEmiR were accounted only (1) if they were identified as DEG in the same brain region where the DEmiRs were identified and (2) if their expression were inversely correlated with each other. This analysis identified that ~71–83% of the total predicted DEG targets were inversely correlated with the DEmiRs (Fig. 8C). Those DEmiRs and DEG targets were used in biological function analyses. Three selected groups (inflammatory, axonal guidance and mitochondrial function) are discussed below.

DEmiRs with inflammatory DEGs

Target scan database identified the DEmiRs inversely correlated inflammatory DEGs in each of the four brain region (Fig. 9B and Supplementary Material, Table S6). There were six common DEmiRs across the four brain regions: miR-1224-3p, miR-1249-3p, miR-155-5p, miR-181c-5p, miR-34b-5p and miR-490-3p, which suggest a universal global effect of these miRs on the inflammatory genes in the 4L;C* brain. The inverse correlation of three of these DEmiRs with their predicated DEG targets is shown in Figure 8C, miR-181C-5p versus *Plau*, miR-34b-5p versus *Gfap* and miR-490-3p versus *Fcgr2a*. Post-IFG treatment, the number of DEmiRs was significantly reduced compared with the untreated mice (Fig. 9B and Supplementary Material, Table S6).

DEmiRs with axonal guidance DEGs

The DEmiRs inversely correlated with the DEGs involved in axonal guidance are shown in Figure 9B and Table 4 and listed in Supplementary Material, Table S6. For instance, miR-10a-5p had increased expression in 4L;C* MID; its target DEGs (*Arhgef11*, *Epha8*) showed decreased expression (Fig. 9D). IFG treatment reduced the number of DEmiRs in this pathway and normalized the expression of some of miRs/target DEGs including miR-10a-5p and *Arhgef11*, *Epha8* (Fig. 9D). Quantitative real-time (qRT)-PCR validated the expression of miR-10a-5p and its mRNA targets (*Arghgef11* and *Epha8*) in MID and supported the RNASeq data (Supplementary Material, Fig. S3).

DEmiRs with mitochondrial DEGs

Twenty-nine DEGs involved in mitochondrial dysfunction across 4 brain regions were inversely correlated with 40 DEmiRs (Table 5). The list of DEmiRs and the target DEGs in each region are tabulated in Supplementary Material, Table S6. Two DEmiRs and their mRNA targets, miR-181c-5p and *Prdx3* and miR-423-5p and *Atp5g3*, showed altered expression and were normalized by IFG treatment (Fig. 9E). The expression of miR-181c-5p/Prdx3 and miR-423-5p/Atp5g3 was validated by qRT-PCR confirming the RNASeq results (Supplementary Material, Fig. S3).

In these analyses, new pathways were identified, i.e. axonal guidance signaling and synaptic transmission that potentially contribute to neuronal phenotype in nGD. Altered gene expression in eIF2 and mTOR signaling pathways and mitochondrial function implicate the cellular compensatory mechanism for

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Table 4. DEmiRs and DEG target in axonal guidance signaling pathway

DEmiR	DEG targets
mmu-let-7c-5p	Epha3, Epha4, Fzd4, Pappa, Sema4g, Srgap3
mmu-miR-101b-3p	Fzd7
mmu-miR-10a-5p	Arhgef11, Epha8, Itsn1, Nrp2, Unc5d
mmu-miR-1197-3p	Ephb4
mmu-miR-1249-3p	Ephb4
mmu-miR-1251-5p	Sema3a
mmu-miR-129-2-3p	Ablim3, Dcc, Robo1
mmu-miR-1298-5p	Bmp7
mmu-miR-133a-3p	Bmp7
mmu-miR-146a-5p	Pappa, Unc5d
mmu-miR-148a-3p	Adam28, Bmp3, Epha8, Nrp1
mmu-miR-15b-5p	Adamts6, Epha7, Pappa, Ptch2
mmu-miR-181c-5p	Nrp1, Sema3b
mmu-miR-1943-5p	Fzd10
mmu-miR-1a-3p	Fzd7
mmu-miR-203-3p	Adamts6
mmu-miR-301a-3p	Ephb4, Wnt1a
mmu-miR-323-5p	Unc5d
mmu-miR-326-3p	Adamts6
mmu-miR-331-3p	Plxna3
mmu-miR-423-5p	Ephb4, Nrp1, Plxnd1
mmu-miR-491-3p	Ptch2
mmu-miR-671-5p	Fzd2, Plxnb1
mmu-miR-98-5p	Epha3, Fzd3, Pappa, Plxnc1, Sema4c, Sema4g, Srgap1, Srgap3

Table 5. DEmiRs and DEG targets in mitochondrial dysfunction pathway

DEmiRs	DEG targets
mmu-miR-101a-3p	Atp5g2, Ndufb5
mmu-miR-101b-3p	Atp5g2
mmu-miR-1224-3p	Ndufa5, Ndufv3, Sdhc
mmu-miR-1224-5p	Maob, Ndufa11, Ndufa2
mmu-miR-125a-5p	Atp5g2, Ndufa2
mmu-miR-138-5p	Cox6b2, Ndufa4l2
mmu-miR-139-5p	Atp5g3, Ndufb5
mmu-miR-141-3p	Sdhc
mmu-miR-146a-5p	Park2
mmu-miR-146b-5p	Park2
mmu-miR-150-5p	Ndufa11, Uqcr11
mmu-miR-15b-5p	Txn2, Atp5g3
mmu-miR-181c-5p	Prdx3
mmu-miR-193-3p	Uqcr10
mmu-miR-1943-5p	Ndufa10, Ndufa11, Ndufa4l2
mmu-miR-199b-5p	Ndufa10
mmu-miR-19b-3p	Lrrk2
mmu-miR-210-5p	Ndufa6, Ndufv3
mmu-miR-29c-3p	Atp5g1, Atp5g3, Ndufs6, Uqcr11
mmu-miR-300-5p	Sdhc
mmu-miR-3084-3p	Ndufa6, Ndufb4, Prdx3
mmu-miR-3085-3p	Atp5c1, Sdhc
mmu-miR-339-5p	Cox6b2, Ndufv3
mmu-miR-34b-5p	Sdhc
mmu-miR-361-5p	Uqcr11
mmu-miR-370-3p	Ndufb4
mmu-miR-377-3p	Atp5l, Ndufa12
mmu-miR-379-5p	Park2
mmu-miR-381-3p	Ndufab1
mmu-miR-383-5p	Prdx3
mmu-miR-410-3p	Sdhd
mmu-miR-423-5p	Atp5g3, Atp5j2, Ndufa11, Ndufa13, Ndufb4, Ndufb7
mmu-miR-433-3p	Cox6b1, Cox8a
mmu-miR-485-5p	Cox6b2, Ndufa6
mmu-miR-486-3p	Maob
mmu-miR-542-3p	Xdh
mmu-miR-671-5p	Ndufb5, Uqcr10
mmu-miR-873-5p	Sdhc
mmu-miR-879-5p	Sdhc
mmu-miR-96-5p	Ndufa4l2

nGD. Notably, this is the first report suggesting the involvement of miRNA in the dysregulation of mRNA in nGD. The analysis identified the DEmiRs with their target DEGs involved in mitochondrial dysfunction, axonal guidance, inflammation and additional pathways where dynamic alterations of miRNAs may have potential roles in neurodegenerative changes in 4L;C* brain. In addition, the transcriptome profile of the 4L;C* brain identified region-specific dysregulation of mRNA and miRNA. The MID, BS and CO were most affected regions.

In this study, selected DEmiRs and DEGs from extensive data were validated by qRT-PCR and immunohistochemistry in supporting the finding by RNAseq. These enriched data provide important pathways for further functional investigation following protein and RNA quantification by immunoblot, immunohistochemistry and qRT-PCR, and functional validation of the targets by silencing or overexpression of gene or miRNA correlated with functional assays. Importantly, this *in silico* transcriptome analysis of nGD mouse brain regions identified targets for further defining regional molecular pathways in nGD pathogenesis.

Discussion

CNS manifestations are prominent in nGD patients with type 2, acute and type 3, subacute neuropathic variants (1,5). Accumulated substrates owing to defective GCase function propagate CNS pathogenesis in nGD. The studies from human patients, animal and cell models have demonstrated that inflammation, mitochondrial dysfunction, disturbed calcium homeostasis, decreased hippocampal LTP, altered autophagy/proteases function and necroptosis are integral to the disease progression (6,10,25,31–33). To understand the underlying molecular changes, mRNA and miRNA expression was investigated using the brains from 4L;C* mice, a mouse model mimicking subacute Gaucher disease (10), and IFG-treated 4L;C* mice that has an attenuated CNS disease phenotype (16). 4L;C* mice showed defective motor function associated with accumulation of substrates (GC/GS), mitochondrial dysfunction, increased activated microglia and loss of neurons (Fig. 1), and reduced hippocampal LTP in the brain (10). Gene expression analysis in 4L;C*, IFG-treated 4L;C* and the WT brains identified abnormality of region-specific expression profile of mRNA and miRNA and new insight of the biological pathways (discussed below) affected in nGD.

Altered axonal guidance signaling in nGD

Axonal guidance signaling pathway is involved in brain connectivity during fetal development and maintenance and repair of the brain through life. Growing evidence has shown the involvement of axonal guidance signaling in PD and Alzheimer's disease (AD) (34–36). Our analyses identified, for the first time, DEGs in axonal guidance signaling pathway in nGD mouse brain. Those genes include receptors for axonal cues (*EphA3*, *Dcc*, *Semag4* and

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Robo1), Rho GTPase (*Arghef11*) that interacts with semaphorins receptor Plexin B to stimulate Rho-dependent signaling, and growth factor, BMP7. The down-regulation of axonal guidance genes can affect synaptic transmission, which could lead to reduced LTP in 4L;C* brain (10).

Our analyses also provided the insight of miRNA in the regulation of axonal guidance genes. This study predicted an inverse correlation of let-7 with DEG targets in axonal guidance signaling pathway including EphA4. The let-7 miRNA has been suggested regulating homeobox transcription factors (*HoxA1* and *HoxB1*) for targeting to the promoter of EphA4, EPhA2 and Eph receptors (37). miR-423 identified in this study has been found in the distal axons of sympathetic neurons (38). Interestingly, most of the DEmiRs determined in untreated 4L;C* brain were not detected in IFG-treated brain, suggesting a therapeutic role of IFG in normalizing the deregulated miRNA. The substantial reduction of both axonal guidance DEGs and DEmiRs in IFG-treated 4L;C* brain infers a protective role of IFG in delaying the progression of nGD pathogenesis. The regulation mechanism on those axonal guidance genes in Gaucher disease remains to be explored.

Molecular basis of defective synaptic transmission/LTP in nGD

Synaptic plasticity is an essential property of the brain, forming basis for learning and memory. LTP is a model for neuronal plasticity (39). Synaptic dysfunction and impaired LTP have been demonstrated in nGD model and animal models of AD, PD and GM2/GD2 synthase knockout (10,40–45). The mechanism of LTP consists of the release of glutamate from presynapses that activates NMDA-glutamate receptors to allow calcium influx, which activate CamK (Ca^{2+}/calmodulin-dependent protein kinase) II and IV involved in calcium entry into postsynaptic neuron, which is essential for maintaining LTP (46–48). RNAseq analysis of 4L;C* brain determined that many genes for synaptic transmission were altered in 4L;C* brain indicating deregulated synaptic transmission in 4L;C* brain. Among them, ryanodine receptors in Ryrs-mediated calcium-induced calcium release provide a major part of the calcium-induced hippocampal LTP and memory processes (49–51). There are three Ryrs that all are expressed in the brain (50). The expression of all three Ryrs were reduced in 4L;C* brain across four regions. In addition, the expression of the voltage-gated calcium channels and binding proteins were also decreased. The down-regulation of these calcium signaling-related genes suggests a disrupted calcium homeostasis in 4L;C* brain, which may affect LTP and synaptic transmission. Thereby, the reduced expression of these key players for LTP in 4L;C* MID (includes hippocampus) could be the molecular basis for suppressed hippocampal LTP in 4L;C* mice brain.

The regulation mechanism of those key players for LTP is not clear. It has been shown that GC mobilizes calcium release through Ryr and enhances calcium release from ER in Gaucher disease fibroblast and neurons (6,52). GC accumulation could be a primary toxic insult affecting LTP. This was supported by the evidences that GC accumulation remained in IFG-treated 4L;C* brain (Table 1) associated with the abnormal levels of *Ryrs*, voltage-gated calcium channels and NMDA-glutamate receptors (*GRIA3*) in 4L;C* brain. Although LTP was not measured in IFG-treated 4L;C* hippocampus, the decreased expression of those key genes post-IFG treatment would predict that LTP suppression could not be corrected by IFG. These results also imply that those genes are primary targets for substrates accumulation. In addition, the decreased expression of *Ryrs* in 4L;C* brain would lead to reduced calcium release from ER, which contrasts with the findings of enhanced calcium releases through Ryrs from *in vitro* studies (6,52). Reduced *Ryr* expression has been reported in AD model (53,54). It is anticipated that reduction of Ryr genes expression is a compensation event as a feedback regulating the calcium homeostasis during disease progression in nGD.

Molecular changes for mitochondrial dysfunction in nGD

Mitochondrial dysfunction is one of the pathological features and demonstrated in nGD cell and animal models (31,32). However, the genes and miRNAs related to mitochondrial function in nGD have not been studied. Using our nGD model, reductions of ATP production and oxygen consumption were seen to be associated with differential mitochondrial mRNA and miRNA expression. The DEGs in the mitochondrial group in 4L;C* brain are greatest in the MID region, suggesting this region is more susceptible to oxidative stress. Those DEGs encode (NADH dehydrogenase, cytochrome *c* oxidase, H+ transporting ATP synthase and ubiquinol-cytochrome *c* reductase) that are components of all five complexes in the mitochondria. Most of them had increased expression. The increased expression of those nuclear-encoded mitochondrial genes may compensate for the mitochondrial dysfunction in the neurodegenerative 4L;C* brain as has been reported in AD models (55).

This study identified DEmiRs and associated DEG targets involved in mitochondrial dysfunction. Among them, miR-210 is postulated to target units of the electron transport chain and tri-carboxylic acid cycle leading to reduction of the mitochondrial metabolism rate (56). miR-181 influences mitochondrial function by targeting to Bcl2 family genes in astrocytes (57). IFG treatment partially corrected abnormalities of those DEmiRs and their target DEGs. Some miRNAs can enter mitochondria to regulate mitochondrial genomic gene(s), e.g. miR-181c versus *Cox1*, leading to electron transport chain complex IV remodeling and mitochondrial dysfunction. Enhanced expression of *Cox1* was inversely correlated with miR-181c levels in 4L;C* MID (data not shown), IFG treatment normalized the expression of Cox1, indicating mitochondrial genomic genes may also play a role in nGD pathology. Our analysis did not point to the mechanism causing the differential expression of those genes but offered the clues of potential involvement of genes and their regional responses to mitochondrial stress in nGD and support the notion of the role of miRNA in regulating nuclear-encoded mitochondria-related gene(s) and mitochondrial genomic genes.

Additional highlights of this analyses included enhanced eIF2 and mTOR signaling pathways and uniform inflammatory response in nGD mice brain. The increased expression of eIF2 and mTOR genes implies a general activation of cellular protein biosynthesis in response to disease condition, e.g. activation of large number of pro-/anti-inflammatory genes, or transcription regulation of the genes involved in neuronal function. The mTOR signaling pathway has been implicated in learning, memory and LTP via phosphorylation of eIF4E-BP1 and S6 in neurons (58–61). In 4L;C* brain regions, the expressions of DEGs involved in LTP and multiple DEGs involved in the synaptic plasticity were decreased, which suggests a transcription inhibition of those genes in the 4L;C* brains. The decreased expression of those genes could affect the LTP and synaptic plasticity, thereby led to the suppression of LTP in 4L;C* brains (10). The cooperative roles of eIF2 and RP genes in mTOR and eIF2 pathways apparently played the roles in nGD.

The analysis of the inflammatory DEGs and DEmiRs demonstrated a relatively uniform inflammatory effect across brain

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regions, implying a final common pathway of neurodegeneration in nGD. Indeed, excess accumulation of GC/GS and their association with reduced population of MAP2$^+$ cells in 4L;C* brain uncovered the mechanism by which brain macrophages gets activated and potentiates neurodegeneration in nGD. In addition to resident microglia response, the results also showed increased incidence of Ly6C+, Ly6G+ and CCR2+ inflammatory subsets of microglial cells in the brain of 4L;C* mice. These data suggest infiltrated monocytes differentiated into the active subsets of microglial cells/macrophages in the brain also contribute to uniform CNS inflammation in nGD.

Including IFG-treated 4L;C* brain samples in this study provides a set of control data for validation of the data analyses of 4L;C* brain. IFG-treated 4L;C* mice had significant attenuated disease progression and reduced inflammatory signals (Table 1) as reported previously (16), because the inhibitory role of IFG towards GCase IFG treatment did not show expected effect *in vivo* on preventing substrates accumulation as shown by *ex vivo* studies, which raised concern for its clinical application (16,17). However, the improved neuronal phenotype in GD model of 4L;C* mice by IFG implies its effect on cellular functions independent of modulating substrate levels by chaperone's role on GCase. Its effect on normalizing molecular changes could serve as control to validate the transcriptome findings in 4L;C* mice. Here, we see significant reduction of the dysregulated DEGs involved in the different biological processes (mitochondrial, axonal guidance, mTOR and eIF2 and inflammation) to normal levels post-treatment confirming these pathways involved in nGD pathogenesis. Furthermore, the reduced number of DEmiRs in the IFG-treated 4L;C* brain suggests that IFG in the treatment process corrects/prevents dysregulation of miRNA expression in nGD.

This is the first study of miRNA expression in Gaucher disease. Growing evidence supports dynamic changes of miRNAs in neurodegenerative diseases, including PD, AD, Huntington's disease and amyotrophic lateral sclerosis (62–66). Mammalian miRNAs have emerged as important post-transcriptional regulators in biological process, particularly in the brain (67,68). DEmiRs were determined in four regions of 4L;C* brain. With IFG treatment, the number of DEmiRs decreased by 50–60% altering the expression profile of a significant number of DEGs regulated by the DEmiRs to their normal levels. Those DEmiRs have potential roles in regulating their target mRNAs in mitochondrial function, axonal guidance, mTOR signaling and inflammation in 4L;C* brain indicating dysregulation of miRNAs that play a pivotal role in the pathogenesis of nGD. This analysis revealed a dynamic change of miRNAs in Gaucher disease and provides the targets to validate their functions in nGD.

These extensive analyses of 4L;C* mice brain provide new insight of the dysregulated mRNAs and miRNAs underlying neuronal pathogenesis in nGD. The transcriptome analyses are hypothesis-driven approaches. Those hypotheses need be tested *in vitro* or *in vivo*. Selected DEmiRs and DEGs from extensive data in this study were validated by qRT-PCR or immunohistochemistry and confirmed RNASeq results. Future studies will be directed to functional validation (RNA level, protein distribution, biological function) of these DEGs and DEmiRs in their biological pathways.

In summary, novel biological pathways, axonal guidance signaling and synaptic transmission/LTP were identified that potentially affect nGD. Significant overlap of the inflammatory functions and pathways was observed across all the regions, indicating uniform inflammatory response across the brain. However, considerable regional preference was evident for some interesting pathways, like mitochondrial dysfunction, axonal guidance, synaptic transmission and LTP. This serves as a platform to begin to understand the factors responsible for the region-specific variation in terms of signaling pathways, drug sensitivity and neurotoxicity in nGD. The *in silico* analyses here identified the miRNAs and target genes that are expected to underlie the neuronal phenotypes and pathophysiology in nGD. These data provide the molecular basis for further investigation of these biological pathways in nGD and new therapeutic targets.

Materials and Methods

Materials

The following were from commercial sources: RNA Later, mirVana™ miRNA Isolation Kit (Ambion, Austin, TX); Affymetrix Mouse Gene 1.0 ST chips and Affymetrix Mouse miRNA 2.0 array (Affymetrix, Santa Clara, CA); Illumina TruSeq RNA Sequencing kit (mRNA Seq) and Illumina HiSeq2000 (Illumina, Inc., San Diego, CA); Gene Spring 12.6.1 from (Agilent Technologies, Inc., Santa Clara, CA); Avadis® NGS software, Version 1.3.0 (Strand Scientific Intelligence, Inc., San Francisco, CA); JMP Genomics 5 (SAS Institute, Inc., Cary, NC) and Ingenuity Pathway Analysis (IPA) (Ingenuity Systems, Mountain View, CA). Antibodies for CX3CR1, Ly6C, Ly6G, CCR2 (Bio legend, San Diego, CA) and MAP2 (Abcam, Cambridge, MA). Liberase Cl (Roche, Indianapolis, IN) and DNase (Sigma, St. Louis, MO). Antibodies for CD45 and CD11b, LSRII flow cytometer (BD Biosciences, San Jose, CA). Mitochondrial Assay Solution, XF96 Extracellular Flux Analyzers (Seahorse Biosciences, North Billerica, MA). Neural Tissue Dissociation Kit P, C tubes and the gentleMACS Dissociator (Miltenyi Biotec, Inc., San Diego, CA). miScript II RT kit and QuantiFast SYBER Green PCR kit (QIAGEN Sciences, Inc., Frederick, MD).

Mice care, treatment, tissues collection and immunohistochemistry

Generation of 4L;C* mice was as described (10). 4L;C* mice have combined V394L/V394L *Gba1* (4L) and saposin C–/– (C*) homozygosity. The strain background of 4L;C* mice was C57BL/6J/129SvEV. The strain- and age-matched WT mice were used as control. All mice were housed under pathogen-free conditions in the barrier animal facility according to IACUC-approved protocol at Cincinnati Children's Hospital Research Foundation (CCHRF).

As described previously, the 4L;C* mice were treated with IFG by consuming 60 mg/kg/d from drinking water (16). The pregnant dames were given drinking water containing IFG (20 mg/kg/d) starting 3–5 days prior to birth and during the weaning period. After weaning, the treated pups were continued on 60 mg/kg/d IFG in drinking water for entire time course.

The brains from the 4L;C*, IFG-treated 4L;C* and WT control cohorts (three mice per cohort) were harvested at the ages of 45 days. Four brain regions were dissected: CO, BS, MID and CB. These regional brain samples were used for lipid (GC and GS) and RNA (e.g. mRNA and miRNA) analyses.

The brain tissues were fixed in 4% paraformaldehyde and processed for frozen blocks. CD68 monoclonal antibody staining was as described (10). The brains fixed in 10% formalin were processed for anti-Map2 antibody staining (1/400, Abcam ab32454). The BenchMark XT IHC/ISH Staining Module (Ventana Medical System, Tucson, AZ) was used for immunohistochemistry studies at CCHMC Pathology Research Core. Tissues sections were counterstained with hematoxylin.

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Cell preparation and flow cytometry

For flow cytometry, whole brains from 4L;C* and WT control mice at 45 days of age were removed aseptically. Single-cell suspensions from brain were obtained using Neural Tissue Dissociation Kit P and the gentle MACS Dissociator following manufacturer's instruction. Viable cells were counted using a Neubauer chamber and trypan blue exclusion.

For the identification of cellular phenotypes in brain tissues, the cells were suspended in PBS containing 1% bovine serum albumin. After incubation (15 min, 4°C) with the blocking antibody 2.4G2 (FcγRIII/I), these cells were used for FACS staining with antibodies to CD45, CD11b, CX3CR1, Ly6C, Ly6G and CCR2 and their corresponding isotypes for microglial cells, as well as MAP2 and their corresponding isotypes for neurons at 4°C for 45 min. Flow cytometry analyses were performed on a LSR II, where microglial cells were gated first by their typical FSC/SSC pattern based on CD45 positivity and double-stained for CX3CR1 and CD11b. Neurons were gated for CD11b and CD45 negativity and positivity for MAP2. A total of 10^5 events were acquired for each cell type, and their percentages were determined by calculating total percentage of $CD45^{int}$ $CD11b^+$ $CX3CR1^+$ microglial cells and $CD11b^-$ $CD45^-$ $MAP2^+$ neurons. $CD45^{int}$ $CX3CR1^+$ and $CD11b^+$ FACS sorted microglial cells (10^5) were also analyzed for their mean fluorescence intensity against the CCR2, Ly6C and Ly6G inflammatory markers. LSRII flow cytometer and FCS Express De Novo software were used to analyze the data.

Brain GC and GS analyses

Isolated CO, BS, MID and CB regions from 4L;C* and WT mice brains were used for glycosphingolipid analyses following extraction with chloroform and methanol (69). GC and GS analyses were carried out by ESI-LC–MS/MS using a Waters Quattro Micro API triple quadrupole mass spectrometer (Milford, MA) interfaced with Acquity UPLC system as described (14). The levels of total GC and GS in each brain region were normalized to mg tissue weight.

Mitochondrial assays

Brain mitochondria were isolated from 4L;C* and WT mice brain as described with modification (31). Mouse brain tissues were digested with trypsin for 30 min on ice followed by homogenization on the gentle MACS Dissociator using Program m_mito_tissues_01 (MACS, Miltenyi Biotec). Homogenates were suspended in 4 ml ice-cold buffer (1.0 mm KCl, 1.0 mM Tri–HCl and 0.1 mM EDTA, pH 8.0) and mixed with 0.67 ml of 2 m sucrose. The suspension was centrifuged at 1300*g* for 5 min to remove nuclei, un-broken cells and large membrane fragments. The supernatant containing mitochondria were pelleted after centrifugation at 9600*g* for 10 min at 4°C. The mitochondria pellet was resuspended in the Storage Buffer (MACS, Miltenyi Biotec). Total mitochondrial protein was determined using Bradford Assay reagent (Bio-Rad).

ATP production and oxygen consumption were determined with isolated mitochondria using XF96 Extracellular Flux Analyzers (Seahorse Biosciences). The mitochondria were diluted in cold 1× MAS (Mitochondrial Assay Solution) and substrate (pyruvate/malate) (Seahorse Biosciences). The mitochondrial suspension (20 µg mitochondrial proteins in 25 µl) was aliquoted into each well while the plate was on ice. The plate was then centrifuged using a swinging bucket microplate adaptor at 2000*g* for 20 min at 4°C. After centrifugation, 155 µl of pre-warmed (37°C) 1× MAS and substrate were added to each well and incubated at 37°C with no CO_2 supplement for 10 min. After calibration of the analyzer, the plate containing mitochondria was sequentially injected with: (A) 25 µl of 8 µm oligomycin, (B) 25 µl of 27 µM cyanide-p-trifluoromethoxyphenylhydrazone and (C) 25 µl of 50 µm antimycin A. Oxygen consumption was measured every 8 min for 16 intervals. The data for ATP production and oxygen consumption rate were analyzed using the XFe Wave software. The respiration parameters were normalized to milligram of mitochondrial protein.

RNA isolation and samples

RNAs in the brain tissues were isolated by being immediately immersed in RNA Later. RNAs (mRNA and miRNA) were extracted using the Ambion mirVana™ miRNA Isolation Kit (Ambion, AM1560) following manufacturer's instructions.

RNA samples were from the four brain regions (CO, CB, MID and BS) in each of three cohorts (WT, 4L;C* and IFG-treated 4L;C*). Each brain region/cohort combination had 3 biological replicates for a total of 36 RNA samples. All RNA samples were sequenced and analyzed for DEGs and differentially expressed miRNAs (DEmiRs). In total, the RNA and miRNA samples sequenced provided 72 data sets for analyses.

Quantitative (q)RT-PCR

First-strand cDNA was synthesized from RNA (1 µg) using miScript II RT kit (Qiagen, 218161) for miRNA cDNA and mRNA cDNA according to the manufacturer's instructions. To detect mRNA, 50–100 ng mRNA cDNA was used. Primers used for amplification of *Atp5g3*, *Prdx3*, *Epha4*, *Sema4g*, *Epha8*, *Arhgef11* and *Gapdh* (internal control) are in Supplementary Material, Table S7. miRNA (50–100 ng) cDNA was used to detect miRNAs. Primers used for amplification of miR-423-5p, miR-181c-5p, let-7-c-5p and miR-10a-5p are shown in Supplementary Material, Table S7. U6 in QuantiFast SYBER Green PCR kit was the internal control for miRNA. QuantiFast SYBER Green PCR kit (Qiagen, 204054) was used for quantification of mRNA and miRNAs on an Applied BioSystems 7500 Real Time PCR machine. Each qRT-PCR reaction consisted of 5 min at 95°C, 40 cycles of 10 s at 95°C, 30 s at 60°C. Expression levels of miRNA and mRNA from qRT-PCR were calculated by the $2^{-\Delta\Delta CT}$ method.

mRNA preparation for mRNASeq

mRNA preparation and whole transcriptome analyses using the Illumina Hi-Seq2000 (RNASeq) were performed at the genetic variation and gene discovery core of CCHRF as described (19).

miRNA preparation for miRNASeq

The miRNASeq studies were conducted using the Illumina platform. Total RNA (1 µg) was prepared using the Illumina TruSeq small RNA kit as per the manufacturer's recommendations, except for the library size selection as described below. Briefly, the RNAs were ligated with different adapters at their 5′ and 3′ ends using T4 RNA ligases 2 and 1, respectively. After reverse transcription from the 3′ adapter, RNAse H treatment and second strand synthesis from the 5′ adapter, the double-stranded cDNAs were amplified by PCR with primers tailed with Illumina flow cell-specific sequences and one of 48 possible six-base indexes to complete the library construction. After pooling samples with each different index in equal molar amounts, the fraction of the total RNA libraries containing the miRNAs was selected by agarose gel electrophoresis using a SageScience Pippin Prep system,

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which can be programed to specifically elute the library fragments with inserts of sizes of 18–25 bases. After quantitation and sizing verification on an Agilent Bioanalyzer, the library pool was diluted to 18 nm and separated into individual molecules by hybridization on an Illumina HiSeq flowcell to a density of 850 k/mm^2, bridge amplified on an Illumina cBot cluster generation instrument and then sequenced on an Illumina HiSeq2000 for 50 cycles in a single direction. After demultiplexing of each library pool according to its barcode sequence, a minimum of 2 m reads per sample were available for analyses of miRNA expression levels.

RNASeq data normalization and analyses

Post Binary Alignment/Map files of RNASeq data were analyzed using Avadis® NGS Version 1.3.0 software. Reads were filtered to remove: (1) duplicate reads, (2) non-primary matched reads and (3) reads with alignment scores of <95. Quantification was performed on the filtered reads against the RefSeq annotation. Post-filtering the read counts dropped between 43% for the mRNA and 70% for the miRNA in the brain regions. PCA and multivariate correlations accessed reproducibility and variability among biological replicates. PCA identified two outlier samples in the RNAseq data. The outliers could significantly contribute to the false-positive number of DEGs and DEmiRs and were thus removed from the downstream DEG and DEmiR analyses.

Data normalization was performed with the DESeq package. This package was used to analyze both RNASeq and miRNASeq datasets. DESeq via R script was performed on the filtered reads using three functions (estimate size factors, estimate dispersions and negative binomial test).

The sequencing depth is estimated by the read count of the gene with the median read count ratio across all genes. The method is based on the negative binomial distribution, which allows for less restrictive variance parameter assumptions than does the Poisson distribution. The false discovery rate (FDR) was calculated according to the Benjamini and Hochberg algorithm (70). FC (±1.5) with an FDR of 0.05 was used as criteria for selection of DEGs and DEmiRs.

miRNA-targeted mRNA prediction

TargetScan database was used to predict the potential DEmiR targets based on the 'seed' sequence of a 6- to 8-mer located on the 3′ UTR of the mRNA (30). Potential regulatory relationships between mRNA and miRNA expression levels were investigated in IPA miRNA target filter by cross-referencing mRNA and miRNA that are differentially expressed, highlighting predicted miRNA and mRNA pair. We identified (1) predicted mRNA target, (2) predicted DEG target and (3) sublist restricted to inversely correlated DEGs. The subset of DEmiRs with inversely correlated DEG targets were further analyzed for their biological function.

Classification of common/unique genes and inflammatory/non-inflammatory genes

The common and unique functional groups associated with the DEGs were evaluated by Venn diagram overlapping the DEGs in each region. The number of inflammatory and non-inflammatory DEGs in each sample set was developed with the inflammatory gene list obtained from the IPA database and the DEG list generated from each sample set by two-way Venn diagrams. The inflammatory gene list was created from the IPA database. The region of the Venn diagram overlapping with the inflammatory gene list and the DEG list was identified as inflammatory DEGs, and the remaining genes were identified as the non-inflammatory DEGs.

Functional classification and ontology of DEGs and DEmiRs

The functional classification of DEGs and DEmiRs from RNASeq platforms in brain regions was identified by IPA. The pathways and networks associated with the DEGs and DEmiRs were constructed based upon the published literature and IPA. DEGs and DEmiRs and their corresponding fold change values were imported into the IPA knowledge base v6.3 (www.ingenuity.com) for functional annotation that summarizes the DEGs associated with top biological functions and canonical pathways. A *P*-value cut-off of 0.05 was used to identify significant functions and pathways.

Cluster analysis of gene expression profiles

Heat maps were generated from hierarchical cluster analysis of the DEGs involved in the mitochondrial function group in the four brain regions. Hierarchical clustering was performed by Ward's method using Euclidean distance metric.

Data storage

The information on the sequenced and partially processed RNA-Seq and miRNASeq datasets has been deposited to the National Center for Biotechnology Information (NCBI)'s GEO database. The accession number to access the dataset is: GSE67375.

Supplementary Material

Supplementary Material is available at *HMG* online.

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